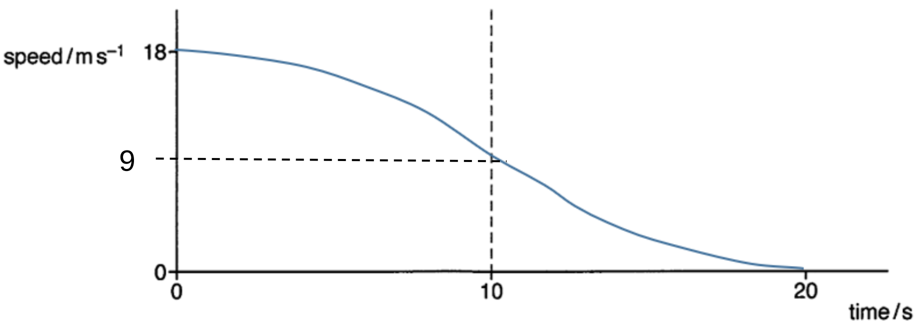
<table>
<tr><td>1(a)(i)</td><td>a = dv/dt
⊗v = + a dt = area enclosed by a-t graph

⊗ v = area enclosed by F-t graph / m (as F = ma)
= [½ (1980)(10)] /1100
= 9.0 m s^{-1}

v = v_i - ⊗v = 18.0 – 9.0 = 9.0 m s^{-1}</td><td>[1 – for ⊗p or ⊗v using area under graph]

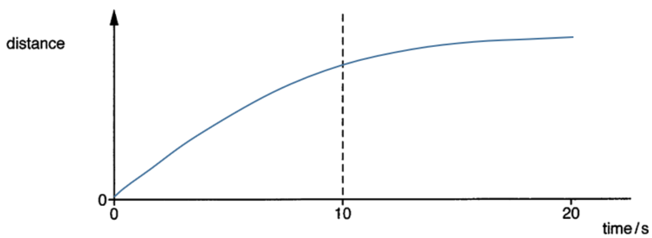
[1]</td></tr>
<tr><td>(ii)</td><td>The net force is the braking force</td><td>[1]</td></tr>
<tr><td>(b)</td><td>

A downward slope with speed starting from 18 m s^{-1} and end at 0

Gradient of curve starts from 0 at t=0 and increases to a max gradient at 10 s.

Gradient of curve decreases gradually from max at 10 s to 0 at t=20 s</td><td>[1]

[1]

[1]</td></tr>
<tr><td>(c)</td><td>

Gradient of the curve decreases non-linearly from a max at t= 0 to 0 at t=20 s.

Distance is 0 at t=0 and max at t=20</td><td>[1]

[1]</td></tr>
<tr><td>2(a)</td><td>sum / total momentum before (interaction) = sum / total momentum after</td><td>[1]</td></tr>
</table>